PHYSICAL CHEMISTRY

GUIDED REVISION PLAN_SCORE ADVANCED [SOLUTION]

Topic : SOLID STATE (Question Bank # 10)

SINGLE CORRECT

1. (C)

Sol. In an ideal ZnS unit cell, 50% Tetrahedral voids are occupied by Zn^{+2} ions

$\Rightarrow$ No. of vacant THV per unit cell $= \frac{50}{100} \times 8 = 4$

$\therefore$ Number of vacant THV in 30 unit cells $= 30 \times 4 = 120$

$\Rightarrow$ Increment in positive charge due to 4 trivalent metal ion is $+4$

In order to maintain electrical neutrality, two more Zn^{+2} ions has to be removed, produces 2 more vacant THV.

$\therefore$ Total vacant THV $= 120 + 2 \Rightarrow 122$

2. (A)

Sol. Given $\frac{\text{Packing fraction}}{\text{density}(\rho)} = 0.1\frac{cm^3}{gm} = \frac{Z \times \frac{4}{3}\pi r^3 \times V_{unit}\ cell}{V_{unit}\ cell \times \text{mass of unit cell in amu.}}$

$\Rightarrow$ $\frac{Z \times \frac{4}{3}\pi\left(10^{-8}cm\right)^3}{Z \times \text{Atomic mass in amu of element}} = 0.1\frac{cm^3}{gm}\Big|_{\text{of unit cell}}^{z = \text{Ranking}}$

$\Rightarrow$ Atomic mass of element $\times \frac{1}{N_A}gm = \frac{4}{3} \times \frac{\pi \times 10^{-24}}{0.1}gm$

$\Rightarrow$ $\frac{4\pi}{3} \times \frac{10^{-24} \times 6 \times 10^{23}}{\times\ 0.1} \Rightarrow 8\pi$

3. (B)

Sol. Given Packing fraction of unit cell $= 0.7 = \frac{Z \times \frac{4\pi r^3}{3}}{a^3}$

$\Rightarrow$ $0.7 = Z \times \frac{4\pi}{3} \times \left(0.2 \times 10^{-7}cm\right)^3 \Big/ a^3$

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$$a^3 = Z \times \frac{4\pi}{3} \times \frac{8\times10^{-24}}{0.7} cm^3$$

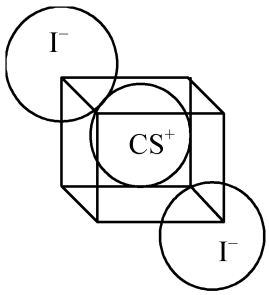
Also Density (ρ) of unit cell = $\frac{\text{mass of unit cell}}{\text{Volume of unit cell}} = \frac{Z \times M(gm)}{a^3 \times N_A}$

$$\Rightarrow \frac{Z \times M(gm)}{Z \times \frac{4\pi}{3} \times \frac{8\times10^{-24}}{0.7} \times 6\times10^{23} cm^3}$$

$$\Rightarrow \quad 3.5 \text{ gm/cm}^3$$

4. **(C)**

Sol.

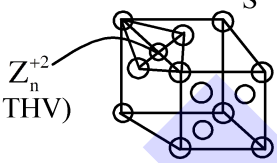


$$r_{CS^+} + r_{I^-} = \frac{\sqrt{3}a}{2} \Rightarrow \frac{1.732 \times 0.445 nm}{2}$$

$$\Rightarrow \quad 0.385 nm$$

5. **(C)**

Sol.



S^{-2} = Every element of face centred cubic site.

Zn^{+2} = Alternate Tetrahedral void

Total nearest neighbour of Zn^{+2}/ C .N of Zn^{+2}

= Total S^{-2} surrounding it = 4

Also S^{-2} occupies vertices of Regular tetrahedron is small cube as shown, as angle, in between two S^{-2} is almost 109.5°

6. **(C)**

Sol. This is a property of a dielectric material, which can behave both as insulator and conductors, depending upon the temperature, $Ti O_3$ (Titan ate) is an example. Also when it is doped with certain elements it can behave as semiconductor.

7. **(D)**

Sol. The oxide of chromium ie Cr(IV) oxide rivals some metals like Mn and Gd and also very good electrical conductors.

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8. **(A)**

Sol. X element at 6 corners $\Rightarrow$ effective no. of $x = 6 \times \frac{1}{8} = 3/4$

Y element at 2 corners and 3 face centres = effective no. of $Y = 2 \times \frac{1}{8} + 3 \times \frac{1}{2} = \frac{7}{4}$

Z element at body centre and 3 face centres = effective no. of $Z = 1 + 3 \times \frac{1}{2} = \frac{5}{2}$

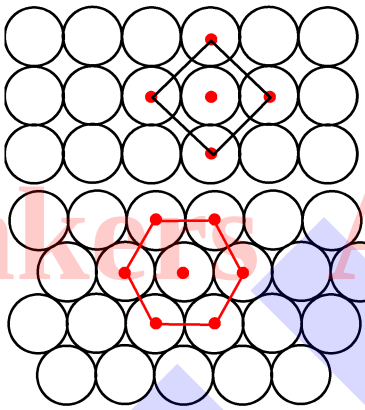
$\Rightarrow$ Density of unit cell $= \frac{\text{mass of unit cell in a.m.u}}{\text{Volume of unit cell in A}^{\circ 3}}$

$\Rightarrow$ $\frac{\left(\frac{3}{4}\times 20 + \frac{7}{4}\times 40 + \frac{5}{2}\times 60\right)\text{amu}}{(2\text{A}^\circ)^3} \Rightarrow \frac{235}{8}\frac{\text{amu}}{\text{A}^{\circ 3}}$

MULTIPLE CORRECT

9. **(C)**

Sol. (A)

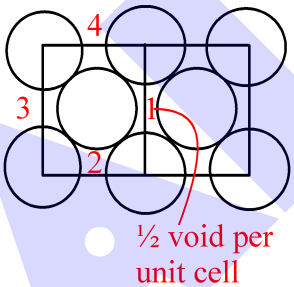


Each fixed sphere touches 4 other identical spheres (CN = 4)

(B)

Each fixed sphere touches 6 other identical sphere (CN = 6)

(C)



Ranking is 2D square close packing

$= 1 + \frac{1}{4} \times 4 = 2$

Total void per unit cell $= 4 \times \frac{1}{2} = 2$ Voids

$\therefore$ void per atom is 1 But in 2D, HCP, Ranking is 3, and 6 voids per unit cell

$\therefore$ Voids per atom is 2

(D) % P.F in 2D, square close packing $= \frac{2\pi r^2}{a^2} \times 100 \Rightarrow \frac{2 \times \pi r^2}{\left(\frac{4r}{\sqrt{2}}\right)^2} \times 100$

$\Rightarrow$ 78.5%

% P.F. in 2D, HCP $= \frac{3 \times \Pi r^2}{6\sqrt{3}r^2} \times 100 \Rightarrow 90.64\%$

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10. **(A,D)**

Sol. (A) Ferrimagnetism is observed when magnetic moments of the domains in substance are aligned in parallel and Antiparallel drections in unequal number. They lose ferrimagnetism and be comes paramagnetic above curies temp.

(B) Opposite statement of that of statement A.

(C) Antiferromagnetic substances like MnO have domain structure similar to ferromagnetic, but their domains are oppositely oriented and cancel out each other's magnetic moment.

(D) They also called permanent magnets. These substance when placed in magnetic fields, all domains get oriented in the direction of magneti field and strong magnetic effect is produced.

11. **(A,C,D)**

Sol. (A) As the ions are missing from its usual site causes decrement in the mass of unit cell, hence density also.

(B) Ions occupies interstitial site, total mass of unit cell remains constant, hence density do not change.

(C) Density may decrease, if an ion is replaced by other ion of lesser atomic mass (total charge must be compensated).

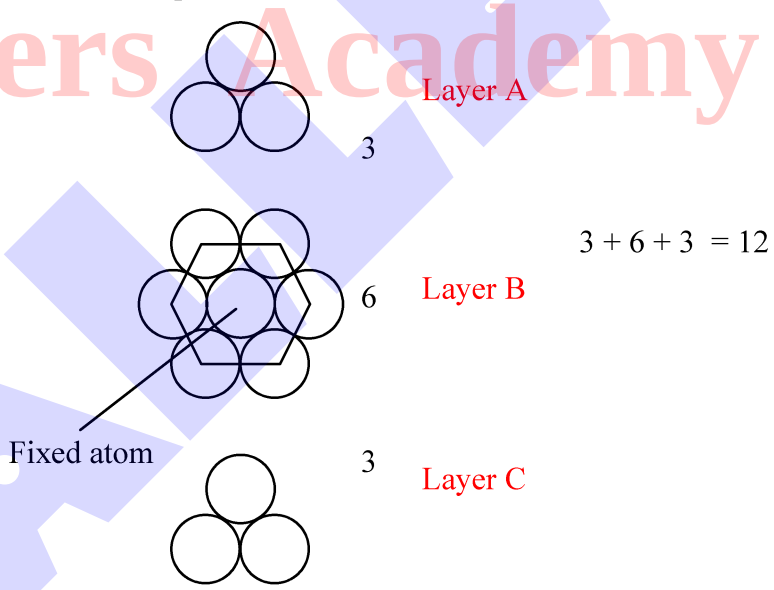
(D) Total mass of unit cell decreases due to ion missing (metallic ion) from its usual site (lost charge is compensated by increment in charge ar neighbour ions).

12. **(B,C)**

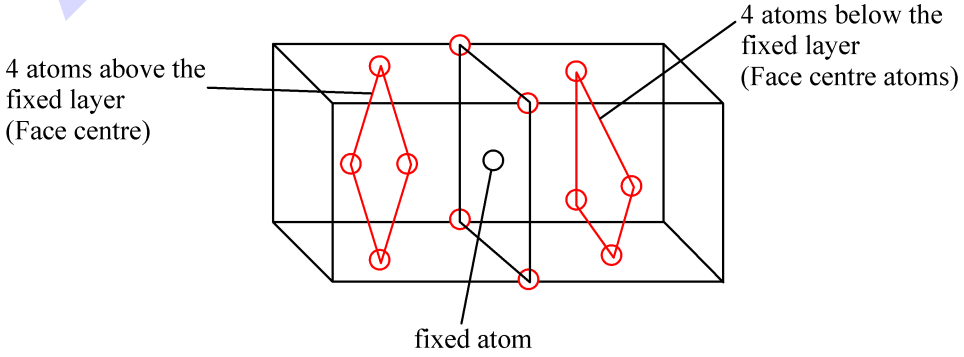
Sol. Fcc structure in 3-D can be represented as

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(B)



(C)



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13. (B)

Sol.

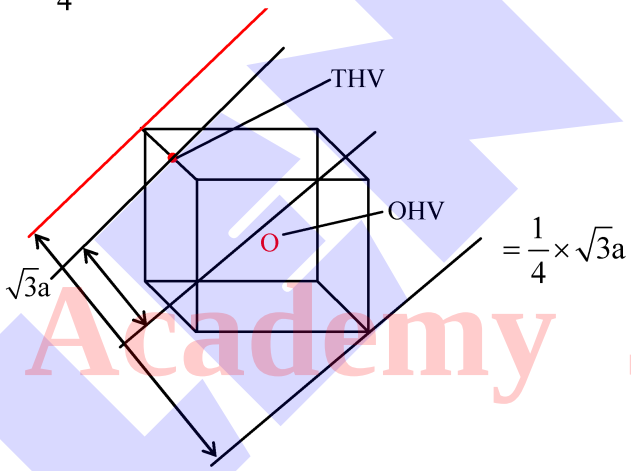
Ions	Position in unit cell	Effective no. per unit cell
O^{-2}	FCC site	4
Fe^{+2}	$\frac{1}{8}$ of THV	$\frac{1}{8}\times 8 = 1$
Fe^{+3}	$\frac{1}{2}$ of OHV	$\frac{1}{2}\times 4 = 2$

Total Fe atoms = 3

$\therefore$ Formula : Fe_3O_4

14. (A)

Sol. Distance between THV and OHV is $\frac{1}{4}$ of Body Diagonal



15. (A)

Sol.

Ionic compound	Position of Cation	Position of Anion
Nacl	All Octahedral	FCC lattice site
Z_nS	Alternate Tetrahedral Void	FCC lattice site
Cscl	Cubic Void	Simple cubic lattice points
Na_2O	All THV	FCC lattice sites

Table type

TABLE OF SYSTEM THAT HAS TO REMEMBER FOR Q.16, 17, 18 :

16. (D) 17. (B) 18. (D)

Sol.

Crystal system	Relation between edge length	Relation between Interfacial Angles	Possible Variation found
(1) Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$	Primitive
(2) orthorhombic	$a \neq b \neq c$	$\alpha = \gamma = \beta = 90°$	Primitive, B.C,F.C end centred
(3) Monoclinic	$a \neq b \neq c$	$\alpha = \beta \neq \gamma$	Primitive end centred
(4) Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma$	Primitive

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INTEGER

1. **(2.00)**

Sol. Density of Ideal unit cell $= \dfrac{Z \times M_{amu}}{\left(500 \times 10^{-12} \times 100\text{cm}\right)^3}$

$$= \frac{4 \times 31.25 \times 1.67 \times 10^{-24}}{(500 \times 10^{-12} \times 100\text{cm})^3}\text{gm}$$

$= 1.67\ \text{gm/cm}^3$ OR $1.67\ \text{gm/ml}$

Decrement in density due to defect $= 1.67 - 1.6075 = 0.0625\ \text{gm/ml}$

$\Rightarrow$ mass missing per ml of unit cell due ot defect $= 0.0625\text{gm}$

OR mass missing per litre (1000 ml) of unit cell $= 62.5\ \text{gm}$

$\therefore$ moles of atom missing $= \dfrac{62.5}{31.25} = 2$ moles

2. **(4.00)**

Sol. Truncated octahedron has = 6 square faces

$x = 24$ 8 Hexagonal faces

$y = 6$ 36 edges

$z = 8$ 24 vertices

$z - \dfrac{x}{y} \Rightarrow 8 - \dfrac{24}{6} = 4$

3. **(6.00)**

Sol. Per mole of NaCl contain 10^{-5} moles of $SrCl_2$.

Each Sr^{+2} replaces $2Na^+$ ion (charge balance), one position occupied by Sr^{+2} it self, finally produces 1 Cation Vacancy

$\therefore$ by 10^{-5} moles of $Sr^{+2} \Rightarrow 10^{-5}$ moles of cation vacancy produces

$10^{-5} \times 6 \times 10^{23}$

$\Rightarrow$ 6×10^{18} Vacancies (x = 6)

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